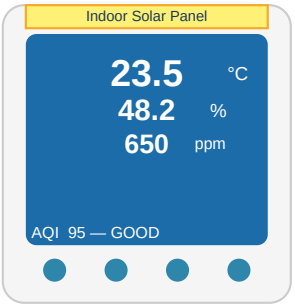
Indoor Ambient Solar-Powered Wireless Environmental Sensor Node

Temperature · Humidity · CO_2 (opt.) · TVOC (opt.) · Ambient Light | Long-Range Wireless | Zero Maintenance | No Battery

The Temco Controls Indoor Ambient Solar-Powered Sensor Node is a completely maintenance-free wireless monitoring solution for commercial and industrial buildings where running cable is impractical or cost-prohibitive.

It collects temperature, humidity, and optionally CO_2, TVOC, and ambient light data, then transmits all readings wirelessly via long-range sub-GHz radio to the paired Temco gateway — which delivers the data to any BACnet or Modbus building automation system over Ethernet or Wi-Fi.

The key innovation is the indoor-tuned solar cell. Standard outdoor solar panels are designed for high-intensity sunlight. This sensor uses a specially engineered cell optimised for the frequency spectrum of indoor LED and fluorescent lighting — so it charges and runs continuously under ordinary office, warehouse, or corridor light with no batteries and no maintenance ever required.



Model: AQ-Solar-LCD
(illustration)

KEY HIGHLIGHTS

- **Indoor-tuned solar cell** — charges under LED and fluorescent lighting; no outdoor sunlight needed
- **Supercapacitor energy storage** — zero batteries, zero maintenance, no shipping restrictions, accepted in all buildings
- **Long-range wireless** — 1 km+ open air, 10+ floors through wood or concrete construction
- **Ad-hoc installation** — mount anywhere indoors that receives ambient light; no cable, no conduit
- Onboard CPU **intelligently reduces reporting rate** during low-light periods, then auto-resumes
- Monitors **temperature, humidity** plus optional **CO_2, TVOC,** and **ambient light**
- Gateway reports over **BACnet and Modbus** — compatible with any standard BMS
- **Dual-interface gateway** — Ethernet and Wi-Fi simultaneously
- **T3000 software is free and open-source** — unlimited sensor trending, SQL logging, floor-plan overlays, email alerts
- Suitable for **hard-to-wire locations**: rooftops, atriums, historical buildings, remote plant rooms

SENSOR SPECIFICATIONS

All data is transmitted wirelessly — there are no wired signal outputs from the sensor node.

Sensor	Range	Accuracy	Interface	Response	Drift/yr
Temperature	-30 to +70 °C	±0.2 °C (5–60 °C)	Long-range wireless	< 10 s	< ±0.1 °C
Relative Humidity	0–100% RH	±2 % (0–90 %)	Long-range wireless	< 10 s	< ±0.5 %
CO■ (optional)	380–32,000 ppm	±(100 ppm + 10%)	Long-range wireless	20 s (τ63%)	< 1 ppm
TVOC (optional)	0–60,000 ppb	IAQ index (calibrated)	Long-range wireless	< 30 s	—
Ambient Light	0–100,000 lux	±10 %	Long-range wireless	< 200 ms	—

** CO_2 and TVOC are optional sensors. TVOC uses a production-calibrated Sensirion SGP sensor and reports an Indoor Air Quality (IAQ) index based on total volatile organic compounds.*

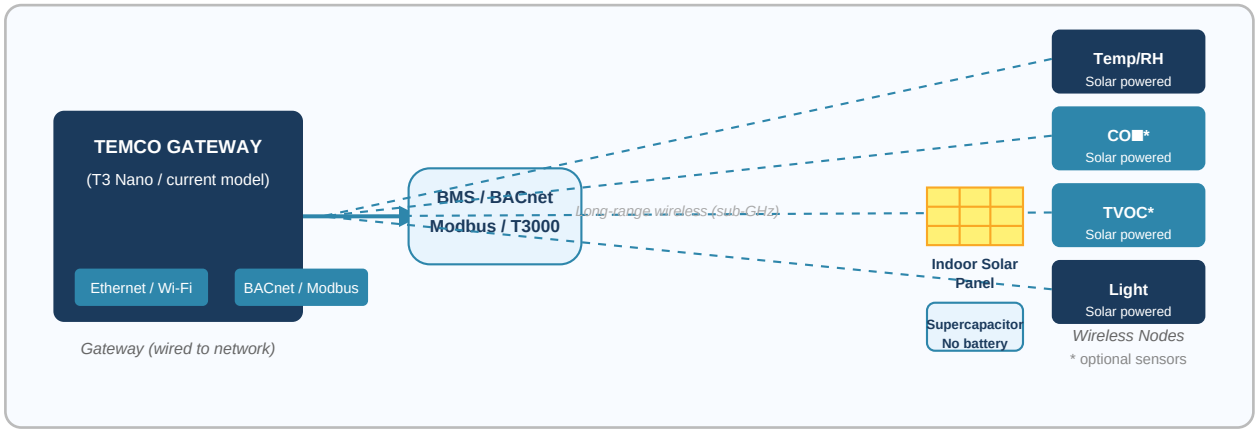
INDOOR SOLAR & SUPERCAPACITOR ENERGY STORAGE

Conventional solar panels target the 400–700 nm visible spectrum at high outdoor irradiance (around 1,000 W/m²). Indoor LED and fluorescent lighting operates at far lower irradiance (100–500 lux) with a different spectral profile. The solar cell in this sensor node is specifically engineered and tuned for these indoor-light frequencies — maximising energy harvest under typical building lighting conditions where a standard solar cell would barely function.

Energy is stored in a **supercapacitor or hybrid supercapacitor** rather than a rechargeable lithium battery. This delivers major practical advantages for building operators:

No battery replacement — ever	Eliminates maintenance visits and associated labour costs
No hazardous materials	No lithium; compliant with shipping and building chemical policies
Building-friendly	Accepted in hospitals, data centres, and facilities with strict battery bans
Intelligent power management	CPU automatically slows reporting rate when stored energy is low, then resumes when light returns
Long service life	Supercapacitors handle far more charge/discharge cycles than batteries

SYSTEM OVERVIEW — WIRING & CONNECTIVITY



The sensor nodes communicate wirelessly with the Temco gateway. The gateway connects to the building network via Ethernet or Wi-Fi and exposes all sensor data over BACnet MS/TP and Modbus RTU/TCP. The sensor nodes and gateway must be used together — they use a proprietary pairing method that ensures reliable, secure communication.

GATEWAY & RADIO SPECIFICATIONS

Parameter	Value	Notes
Wireless Protocol	Long-Range (LoRa)	Sub-GHz; penetrates walls & floors
TX Power	Up to 22 dBm	Configurable per installation
Range — Open Air	> 1 km	Line of sight
Range — Through Floors	10+ floors typical	Wood/concrete; rebar may reduce range
Gateway Connectivity	Ethernet + Wi-Fi	Dual interface, both simultaneous
Protocols Supported	BACnet, Modbus	Standard building automation protocols
Max Sensor Nodes	Contact Temco	Multi-node support confirmed
Operating Temp	-40 to +85 °C	Electronics rating
Supply Voltage	3.0 – 3.6 V (typ. 3.3 V)	Internal regulated

Parameter	Value	Notes
Energy Storage	Supercapacitor / Hybrid	No battery — zero maintenance
Darkness Hold-up	TBD — confirm w/ test	Depends on reporting interval & cap size

HOW IT WORKS — TYPICAL OPERATION SEQUENCE

#	Operation Step
1	Indoor ambient light charges the supercapacitor — no battery required
2	Sensor node stays in deep low-power mode to maximise hold-up time in darkness
3	At the configured interval, the node wakes up automatically
4	Active sensors are powered on; temperature, humidity, CO■*, TVOC*, and light data collected
5	Data packet is transmitted wirelessly via long-range radio
6	Temco gateway receives the packet and maps values to its input registers
7	Data is available over BACnet, Modbus, Ethernet, and Wi-Fi
8	T3000 logs readings to SQL database; email alerts can be configured for any threshold

The gateway maps all received sensor values to its input registers. These are exposed over BACnet (standard in HVAC / commercial buildings) and Modbus (standard in industrial systems) to any connected building management system. T3000 can log an unlimited number of sensor readings for an unlimited duration — storage is only limited by hard-drive capacity.

T3000 SOFTWARE — FREE & OPEN SOURCE

Free download, no licence fee	No subscription, no per-device cost
Live dashboard	View all sensor values in real time across the building
SQL trend logging	Handles millions of records; unlimited sensors, unlimited history
Floor-plan views	Place sensor icons on building drawings and see live readings
Email & push alerts	Notify staff automatically when CO_2, temperature, or humidity exceed setpoints
BACnet & Modbus integration	Connect to any third-party BMS platform with no extra middleware
Open source	Community-supported; full access to configuration and scripting

INSTALLATION & COMMISSIONING

Installation is straightforward — mount the sensor wherever it receives consistent indoor light, then pair it with the Temco gateway using the firmware pairing procedure.

- Mount the sensor node where indoor ambient light is present — ceiling, wall, or column
- Avoid locations in permanent shadow unless a larger energy storage option is specified
- Power the Temco gateway and connect it to the building network via Ethernet or Wi-Fi
- Pair the sensor node to the gateway using the firmware pairing procedure (see gateway setup guide)
- Confirm live values appear in T3000 before enabling trend log recording
- For commissioning validation: log supercapacitor voltage, reporting interval, and packet reception quality over 24–48 hours

Note: Sensor configuration, reporting interval, and power behaviour may vary by firmware version. Solar performance depends on installation location and available light intensity. Contact Temco Controls for current firmware documentation and gateway model information.

ORDERING INFORMATION

Contact Temco Controls for current part numbers, gateway model designation, optional sensor configurations, and volume pricing. The sensor node and gateway must be ordered and used together.

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